

Simulating Neutron Transport and Key Behaviours in Heterogeneous Fusion Breeder Blankets

MPhys Research Project

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Abstract

The continuous production of tritium is a pressing issue in deuterium-tritium (D-T) fusion research. Breeder blankets offer a potential solution. Consisting of lithium-based compounds, breeder blankets produce tritium upon interacting with fusion neutrons. Building them into the reactor structure allows for in-situ tritium breeding while also providing shielding and heat extraction. Maintaining tritium self-sufficiency is a key goal for practical breeder blankets. The goal of this project is to use a Monte Carlo simulation of neutron transport to evaluate and compare the tritium production capabilities of several breeder blanket designs. In this project we carried out systematic parameter investigations to characterise the tritium breeding behaviour of a variety of blanket structures and compositions. By simulating a homogenous blanket structure we determined that a water-cooled, liquid $\text{Li}_{17}\text{Pb}_{83}$ breeder had the highest Tritium Breeding Ratio (TBR) of the materials under investigation with a value of 1.61. The maximum TBR was obtained for a breeder module thickness of $\sim 500\text{mm}$, a molar lithium-6 enrichment of 40% and a zirconium alloy first wall. Overall, the observed trends matched the behaviours we expected from fusion neutron and produced results that demonstrated good agreement with the available literature.

1 Introduction

Deuterium-Tritium (D-T) fusion has long been considered to be the optimal reaction for use in a commercial fusion reactor (Heckrotte and Hiskes, 1971). Its large reaction cross section and energy output are key factors in its selection. Unfortunately, it has a key downside. Tritium is a radioactive isotope with a half life of 12.32 years so it only exists in trace amounts on earth. Consequently, there is a significant body of work focussed on the reliable breeding of tritium for use in fusion reactors. Instead of relying on external tritium production, many reactor designs include a breeder blanket. The reactor vessel is surrounded with lithium which when bombarded with fusion neutrons releases tritium as shown in the equations below.



The primary reaction involved in tritium breeding involves ${}^6\text{Li}$ and produces one consumes one neutron to produce one triton. Clearly this one-to-one neutron economy is not sustainable if any parasitic absorption takes place. Thus, to guarantee reliable tritium production greater than the threshold of equilibrium requires the production of additional neutrons. Elements such as lead or beryllium undergo a multiplication (n,2n) reaction upon interaction with neutrons and so are mixed into the breeder material.

With many prototype reactors planned for construction by the 2040's (Barbarino, 2018) there is significant interest in developing and refining the principles of tritium breeding. Breeder structures are evaluated in terms of their Tritium Breeding Ratio (TBR) using Monte Carlo simulations of neutron transport. A TBR of greater than 1 is necessary to maintain equilibrium. In the first half of our project (Khan, 2025; Peters, 2025) we performed neutronics simulations on a model blanket for a variety of materials, enrichments and blanket thicknesses. A water cooled blanket consisting of a eutectic lithium-lead alloy ($\text{Li}_{17}\text{Pb}_{83}$) produced the best TBR results. However, our investigation had significant limitations. Unlike practical blanket designs which separate the breeder and coolant into a series of distinct, alternating layers (Lee et al., 2012; Qu et al., 2020; Boullon, Jaboulay and Aubert, 2021) our simulation used a homogeneous mixture of the two. Additionally, we used a simplified ring neutron source instead of a more realistic plasma body source. By utilising more complex blanket designs we hope to gain a deeper understanding of their performance requirements. With this in mind, it is necessary to carry out investigations beyond simple measurements of TBR and to consider other factors that may favour or rule out certain configurations. For example, the effect of neutron flux on the behaviour of key structural components throughout the lifetime of the blanket.

Another flaw in our implementation during the homogeneous blanket studies was a lack of multidimensional investigation. Simulations were carried out by varying individual parameters making it difficult to isolate the primary influences on the behaviour of TBR. This is made even more difficult and time consuming as heterogeneous effects are introduced. A Machine Learning (ML) programme is beneficial in this regard (Zhou, 2021), allowing for the investigation of complex, multidimensional variables.

2 Theory

2.1 Heterogeneity in Blanket Designs

Heterogeneous neutron transport simulations can exhibit radically different behaviour to simpler, homogeneous models. This has long been understood in nuclear fission [REFERENCE?](#) where separating the fuel and moderator leads to a higher reactivity due to more efficient moderation. It is reasonable to assume that the neutron economy of a fusion reactor will also be influenced by heterogeneity. Investigations by Lee et al., 2012 and Qu et al., 2020 suggest that a heterogeneous breeder blanket would have a reduced TBR compared to a homogeneous simulation.

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2.2 Material Damage

In addition to evaluating the TBR of heterogeneous blanket structures we also recorded several other tallies including the damage inflicted on the structural components. When irradiated with neutrons the atoms in a material are displaced from their position in the lattice. The number of displacements (N_d) depends on the the damage energy delivered to the material (T_d) and the threshold displacement energy (E_d).

$$N_d = \frac{0.8T_d}{2E_d} \left((1 - c) \left(\frac{0.8T_d}{2E_d} \right)^b + c \right) \quad (3)$$

Where b and c are material constants that express the recombination rate of displaced atoms (Saha, Devan and Ganesan, 2019). Extrapolating this value to full breeder structures requires conversion to displacements per atom (dpa). In a reactor operating at power P the dpa per full-power-year can be expressed as

$$dpa \text{ fpy}^{-1} = \frac{N_d}{\left(\frac{\rho V N_A}{\mu} \right)} \cdot \frac{P}{E_f} \cdot (365.25 \cdot 24 \cdot 60^2) \quad (4)$$

for a structure of volume V , density ρ and molar mass μ . The energy released during fusion is represented by E_f .

High-energy neutron flux has a wide variety of effects on irradiated materials. A critical one for fusion materials is helium induced embrittlement. Helium produced in transmutation reactions accumulates at grain boundries, causing swelling and weakening the structure (Trinkaas, 1985). It is possible to express critical helium concentrations as a lifetime dpa limit for selected materials (Gilbert et al., 2012).

2.3 Machine Learning

The tritium breeding behaviour of heterogeneous breeder blankets is a complex and multivariate problem. While it is possible to isolate trends by manually investigating a wide range of blanket configurations this process is incredibly time consuming and it is easy to miss important results. Instead, many neutronic simulations rely on Machine Learning (ML) algorithms to optimise tritium breeding (Barker, Gilbert and Edmondson, 2026; Humphrey et al., 2026).

Bayesian optimisation is the standard method applied in these circumstances (Shahriari et al., 2016). Essentially, an initial guess is updated with new data to produce a better prediction.

Prior to the optimisation process a set of datapoints taken at random positions in the model space is generated. These values are used to produce a surrogate model, a function that aims to mimic the behaviour of the simulation. Instead of modelling $\sim 10^6$ neutrons for every possible datapoint a surrogate is defined based on the mean (μ) and variance (σ) of each datapoint in the set. Gaussian Process Regression (GPR) is a popular surrogate in neutronics modelling since it requires a relatively small sample size and can easily account for stochastic noise resulting from the probabilistic nature of neutron transport (Polke, Ahle and Söfker, 2026).

Once a surrogate has been created it can be optimised. This process is driven by an acquisition function. A mechanism that selects where in the model space to carry out the next investigation. The Upper Confidence Bound (UCB) method makes the selection based on the mean and variance of the surrogate function (Mia et al., 2025; Ribeiro et al., 2026).

$$\alpha = \mu(x) + \sqrt{\beta}\sigma(x) \quad (5)$$

Higher values of β prioritise the investigation of areas with high uncertainties. The process of testing, acquisition and simulating is repeated until the surrogate model converges on the optimised value.

3 Methodology

3.1 Blanket Structure

Many heterogeneous breeder blankets are designed with several ring-like layers, with alternating breeder material and coolant separated by structural material. There is significant variability in design, ranging from only two layers (Gohar et al., 1998; Segantin et al., 2020) to five (Lee et al., 2012; Qu et al., 2020). A four layer configuration (as demonstrated in Fradera et al., 2021) was selected for this investigation. As depicted in figure 1b, the blanket is shielded by a 25mm thick first wall consisting of Zircaloy-4 (King et al., 2022). The available literature supports the placement of the breeder as the nextmost layer before the coolant (Klix et al., 2005; Boullon, Jaboulay and Aubert, 2021). An eutectic $\text{Li}_{17}\text{Pb}_{83}$ breeder and water coolant were selected with thicknesses of 80mm and 30mm respectively. The lithium in the breeder layers was enriched to 40% Li_6 . Each layer was separated by a 10mm of Eurofer97 (Stornelli et al., 2024). The composition and structure of each blanket layer was chosen for its neutronic properties demonstrated in the homogeneous blanket simulations (Peters, 2025).

To comprehensively investigate the effects of heterogeneity on TBR it was important that we attempt to explore the effects of modifying individual layer geometry. Unfortunately, it would be too time consuming to simulate each individually. Instead, a factor denoted by ζ was used to impose a linear gradient on the thicknesses of each layer. Each breeder and coolant pair had their thickness altered by a factor of $1-2\zeta$, $1-\zeta$, $1+\zeta$ and $1+2\zeta$ sequentially. Thus, negative values of ζ led to thicker front layers (figure 1a) and positive values led to thicker rear layers (figure 1c). Implementing this gradient meant that the relative impact of breeder location could be studied more effectively. It is important to note that thicknesses of the first wall and separators were kept constant throughout.

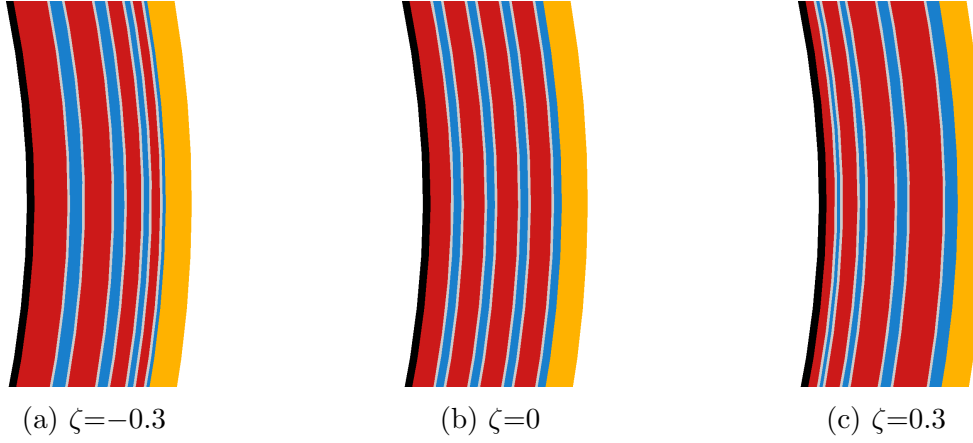


Figure 1: 2D cross sectional diagrams of the breeder blanket layers, beginning with the first wall (black) and then alternating between breeder (red) and coolant (blue) layers separated by structural layers (grey). The rear containing wall is highlighted in yellow. Each slice shows the relative thicknesses of the layers for values of ζ ($-0.3, 0, 0.3$ from left to right).

3.2 Simulation Tallies

The tally which holds the most importance in a majority of breeder simulations is TBR. It provides a simple method of comparison, focussing on the main goal of breeder blankets which is the production of tritium. A value of ≥ 1.05 is typically selected as a threshold for sustainable tritium breeding. However, there are many other factors that influence the selection of breeder materials and dimensions. OpenMC provides several tallies that can be used to probe the full functionality of the blanket and investigate the mechanics of tritium breeding in more detail (Romano et al., 2015).

To achieve a TBR greater than equilibrium it is vital that sufficient neutron multiplication reactions take place. The OpenMC 'multiplication' tally counts the number of $(n, 2n)$ reactions that take place per source neutron. We would expect a vast majority of these reactions to take place within the breeder layers containing the lead multiplier. When neutrons interact with nuclei they are either absorbed or scattered. The rate at which these reactions take place per source neutron is measured by the 'absorption' and 'Scattering' tally respectively. In a simulation with minimal leakage we would expect the total absorption to be equal to the one plus the multiplication value. Since neutrons can be scattered several times before they are absorbed the 'scattering' tally should record much higher values. During these interactions neutrons also impart energy into the materials, measured in eV per source neutron by the 'Heating' tally. The energy that results in atomic displacement from the crystal lattice is logged by 'damage-energy'. Finally, the 'flux' tally was used to measure the neutron flux through the blanket. OpenMC calculates flux by tracking neutron paths in units of particle-cm per source neutron. Integrating over the volume provides us with a value for the fluence (neutrons/cm²).

3.3 Machine Learning Implementation

To gain a full understanding of the blanket behaviour it became necessary to utilise a machine learning framework. For this we used BoTorch (Balandat et al., 2019), an ML algorithm specialising in Bayesian optimisation. We began by carrying out separate 4-dimensional investigations, optimising TBR for the thicknesses of the four breeder layers then the four coolant

layers. Then we carried out an 8-dimensional simulation of all layer thicknesses. In all cases the workflow was essentially the same. A set of 50 TBR values were generated using random points in the parameter space. Gaussian Process Regression was used to create a surrogate model using these values. An acquisition function defined by the UCB method (equation 5) was then used to select a configuration to explore. A value of $\beta=1,000,000$ was selected to ensure that the model prioritised the exploration of areas with high uncertainty in the parameter space. The blanket defined by these parameters was simulated and the TBR added to the dataset, modifying the surrogate model. This process was repeated 50 times to evaluate a blanket configuration that resulted in optimal tritium breeding.

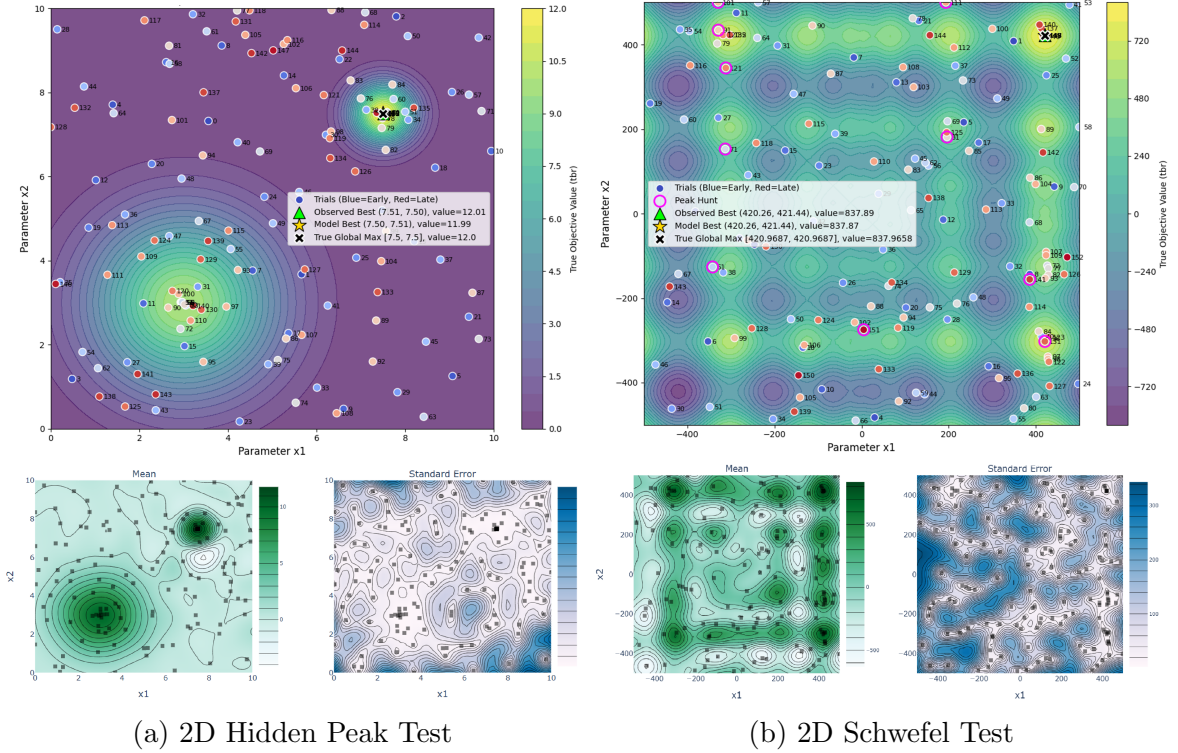


Figure 2

To guarantee the validity of our machine learning programme it was necessary to carry out a series of low dimensionality test prior to full implementation. The results of two example tests are shown in figure 2. Figure 2a depicts the results for a 2D hidden peak test. A short, wide peak and a tall, narrow peak are present which tests the algorithms ability to locate narrow maxima without getting caught on local hills. Meanwhile, figure 2b outlines the results of a Schwefel function test (Schwefel, 1981), where there are several similar peaks spread throughout the grid. This function aims to evaluate the programmes ability to comprehensively explore the parameter space when there are many candidate peaks. In both examples the our ML implementation isolated the true peak and produced an acceptable surrogate model, as seen beneath the main plots.

4 Results

4.1 Heterogeneous Blanket Results

4.1.1 Heterogeneous Effects

We began with a straightforward investigation into the effect of introducing heterogeneity into a blanket on its tritium breeding capabilities. TBR was recorded at a range of blanket thicknesses for homogeneous and heterogeneous configurations of a water cooled, $\text{Li}_{17}\text{Pb}_{83}$ breeder. Shown in figure 3, the heterogeneous design has a significantly lower TBR, not very much higher than equilibrium. A simulation was also carried out for a homogeneous blanket with a fraction of Eurofer97 proportional to that found in the separator layers of the heterogeneous blanket. It demonstrates that while blanket composition does play a role in the reduced TBR, there are clearly heterogeneous effects at play.

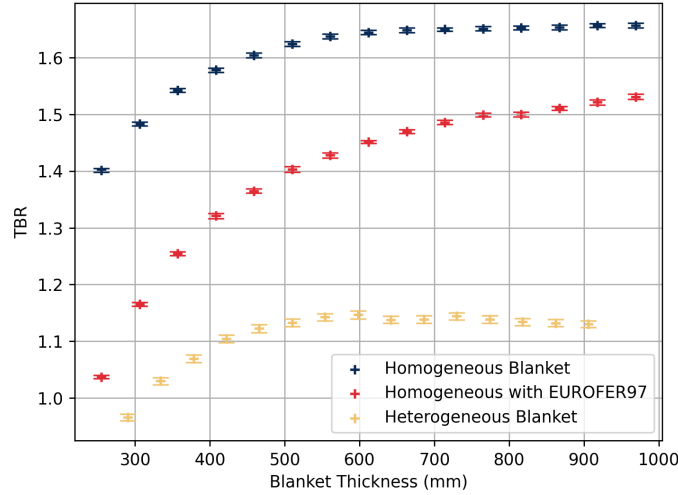


Figure 3: A graph showing the TBR at various thicknesses comparing the results for a homogeneous blanket (Peters, 2025) and the heterogeneous configuration in figure 1. Data was also gathered for a homogeneous blanket with a proportional mixture of structural Eurofer97.

Figure 4 depicts the contribution of each breeder layer to the total TBR at various total thicknesses. The thickness breeder and coolant layer was increased proportionally so that the relative size and coolant fraction remained constant. Breeder layer 1 was situated closest to the neutron source with breeder layer 4 furthest away. The fore layers yielded the highest TBR with layer 2 generating the most in thin blankets and layer 1 producing more with increasing thickness.

4.1.2 Plasma Body Neutron Source

In addition to modifying the blanket structure, the ring neutron source used in the previous semester was swapped for a more realistic plasma body source. The parameters used to define the plasma body are specified in Fausser et al., 2012. TBR is plotted against triangularity for a heterogeneous blanket employing both neutron sources in figure 5. For both sources the TBR and triangularity show no discernable trend. The ring source results in a roughly 2.5% greater tritium breeding rate than the body source. **WHY IS THIS. LESS ENERGY AND FLUX. WHY**

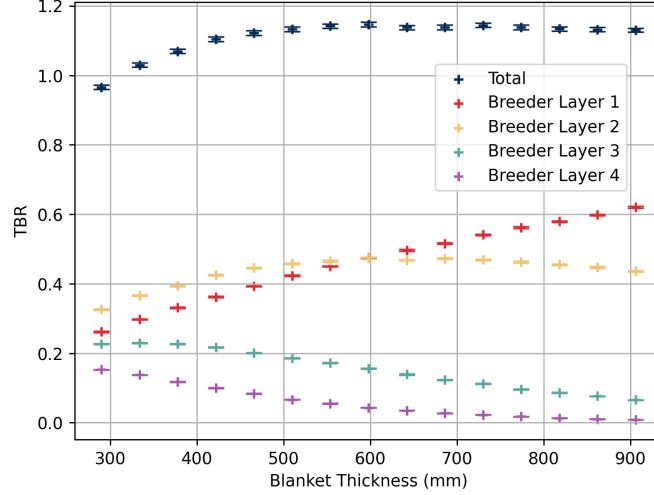


Figure 4: A plot of the TBR contributions of each breeder layer and the total for range of blanket thicknesses.

4.2 Linear Gradient Results

4.2.1 TBR

Once it was established that heterogeneity resulted in a discernable reduction in TBR it became necessary to probe this behaviour and deepen our understanding of the breeder behaviour. The linear gradient parameter ζ proved useful in this regard, allowing for the investigation of effects in the front and rear of the blanket. It is worth noting that all tallies are given in units per source neutron. The leftmost graph of figure 6 exhibits the TBR contribution from different breeder layers at different ζ values. For more frontloaded designs (negative values), the first layer dominates however at $\zeta > 0$ the second layer produces more tritium. Layers 3 and 4 contribute comparatively little, except at large, positive ζ .

On the right of figure 6 a heat map shows the tritium breeding reactions taking place throughout the structure for $\zeta = 0$. As we would expect almost the entirety of tritium breeding reactions occur in the breeder layers, decreasing with depth into the blanket.

4.2.2 Multiplication

As previously stated, neutron multiplication reactions ($n,2n$) are necessary to ensure a TBR greater than equilibrium. The graphs in figure 7 depict the contribution from each layer type on the left and a heatmap of multiplication density on the right. The majority of multiplication occurs in the breeder layers where neutrons interact with lead nuclei. However, there is a small contribution from the zirconium in the first wall and the iron in the separators. With increasing ζ the multiplication in the breeder layers and the blanket overall declines steadily, despite a small increase within the structural layers. Almost the entirety of the multiplication takes place in first and second layers, as shown in the heat map.

4.2.3 Absorption

To maintain a positive neutron economy it is vital that parasitic neutron absorption is kept to a minimum. On the right of figure 8 the absorption rate of each layer is plotted against ζ where the total declines steadily. The heat map on the right shows the absorption reaction

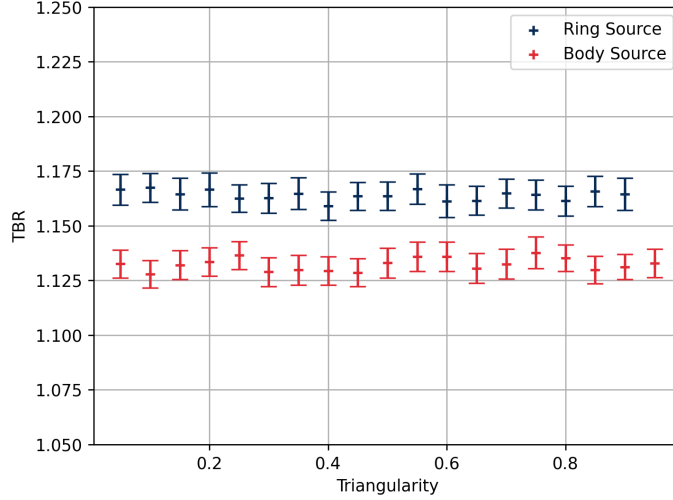


Figure 5: The TBR at various triangularity values for a heterogeneous blanket using a ring neutron source and a plasma body neutron source.

density, decreasing throughout the depth of the blanket. Since the absorption tally also counts tritium breeding reactions, the breeder layer has the highest recorded values, followed by the Eurofer97 separators.

4.2.4 Scattering

In addition to absorption, neutrons can also be scattered when interacting with nuclei, often losing energy in the process. Since a single neutron can be scattered many times the reaction rates recorded in figure 9 are much higher than for previous tallies. As ζ increases the total scattering declines before rising slightly. Scattering primarily takes place in the breeder as a result of it's high density and in the coolant layers due to the high scattering cross section of hydrogen.

4.2.5 Neutron Heating

Upon absorption or scattering neutrons may impart energy into the nuclei, heating the surrounding material. The leftmost graph of figure 10 demonstrates that heating energy remains roughly stable over ζ . Heating energy is overwhelmingly produced in the breeder layers with only a small contribution from the coolant layers and very little else besides.

4.2.6 Damage Energy

OpenMC also allows the damage energy imparted to a material to be recorded. Equations 3 and 4 can be used to convert damage/displacement energy into an expression of dpa per full-power-year. An assumption of a reactor power of 500MW was made based on the planned output of ITER (Aymar, Barabaschi and Shimomura, 2002). Since both the coolant and breeder are constantly circulating liquids it is only necessary to investigate the damage to the first wall and structural layers. A graph plotting the dpa fpy^{-1} against ζ can be seen on the left of figure 11, with a heat map of damage energy density on the right. As we might expect first wall is subjected to a displacement rate much higher than the rest of the reactor structure however, this value declines for rearloaded (high ζ) configurations.

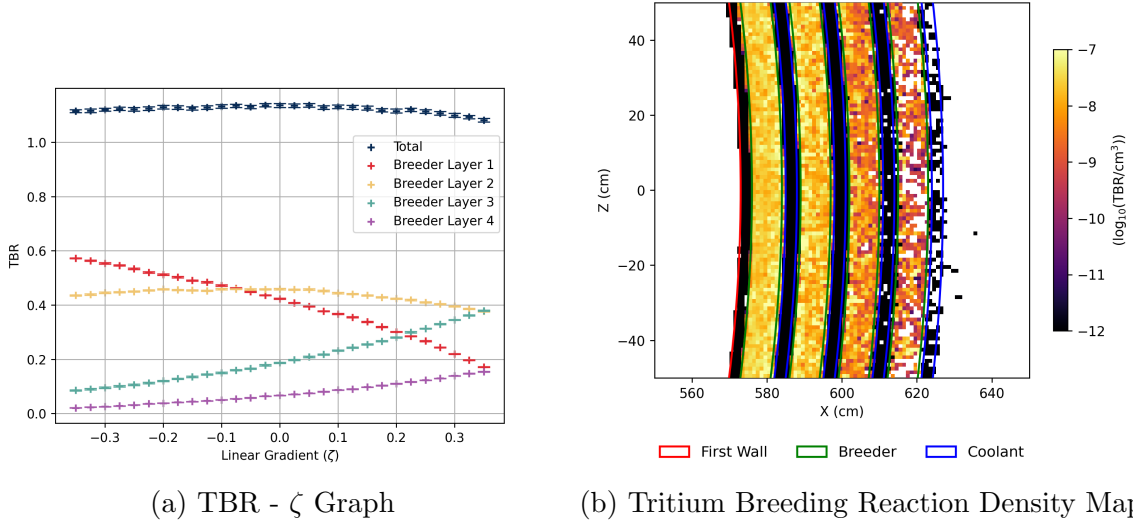


Figure 6: This figure contains two graphs. On the left, the TBR contribution of each breeder layer is plotted for a range of ζ values. On the right, tritium breeding reaction density is shown for a 2D slice of the breeder at $\zeta = 0$.

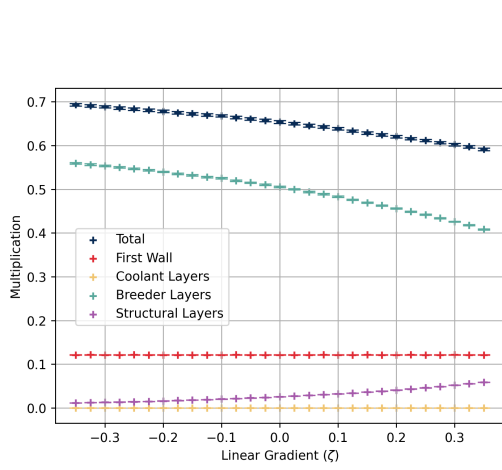
4.2.7 Neutron Flux

Flux is calculated in OpenMC by tracking the neutron paths through a volume and returning a measurement in neutron-cm. Dividing by the volume of the element provides a fluence (time integrated flux). As exhibited in the left graph of figure 12 the breeder layers experience the largest integrated fluence due to their large volume. The heatmap on the right shows that the fluence decreases exponentially at increasing blanket depth.

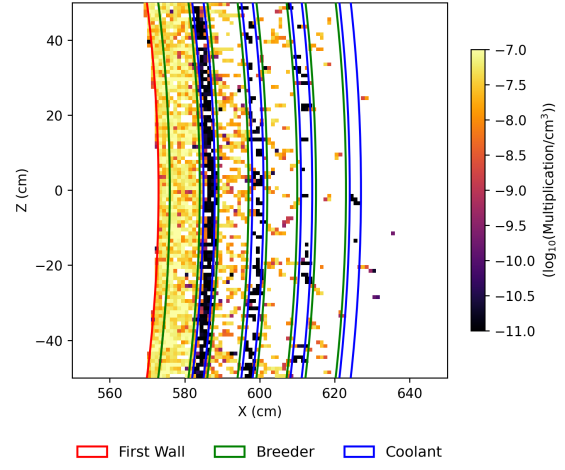
4.3 Machine Learning Results

In addition to the manual investigation of blanket behaviour an ML framework was used to optimise the breeder geometry. The thicknesses of each breeder and coolant layer served as variables that would be explored to evaluate a peak TBR. The breeder layers were limited to between 30mm and 160mm and the coolant layers between 15mm and 60mm. Optimum configurations and the resulting TBR values are displayed in figure 13. Breeder layers are labelled B1 - B4 and coolant layers C1 - C4, counting radially outwards.

To begin with, 4D simulations were carried out where only the breeder layers or only coolant layers were allowed to vary, leading to TBR values of 1.2805 ± 0.0026 and 1.2080 ± 0.0011 respectively. When varying the breeder layers, their total thickness was capped at 320mm while the coolant sizes were fixed at 150mm. Meanwhile in the coolant only simulation their collective thickness was set to 120mm with 80mm breeder layers. Once completed, a series of 8-dimensional investigations were performed where all layers could be modified. Two constrained 8D investigations were performed, both limited to a maximum thickness of 440mm not including separators. An 'extremetised' model which prioritised front loading and a 'decaying' configuration where the breeder thicknesses decreased exponentially. These models resulted in very similar, frontloaded blanket configurations with TBR's of 1.3303 ± 0.0033 and 1.3304 ± 0.0032 respectively. The final simulation removed all constraints on thickness so as to produce a maximum TBR of 1.4055 ± 0.0033 .

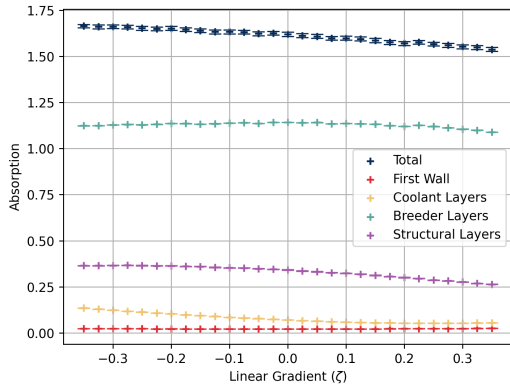


(a) Multiplication - ζ Graph

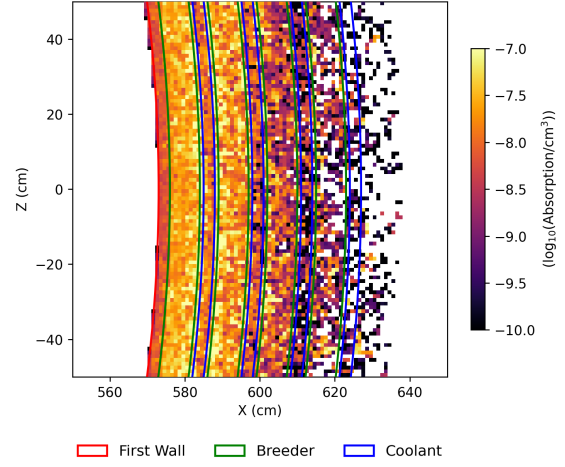


(b) Multiplication Reaction Density Map

Figure 7: The graph on the left displays the neutron multiplication produced in the blanket at different values of ζ . On the left a 2D visualisation of the muliplication density is shown.



(a) Absorption - ζ Graph



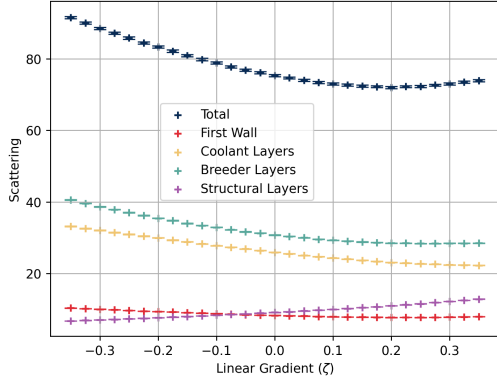
(b) Absorption Reaction Density Map

Figure 8: Two graphs showing the absorption measured in the breeder blanket. A graph of absorption at several ζ values is shown on the left with a 2D heat map at $\zeta = 0$ on the right.

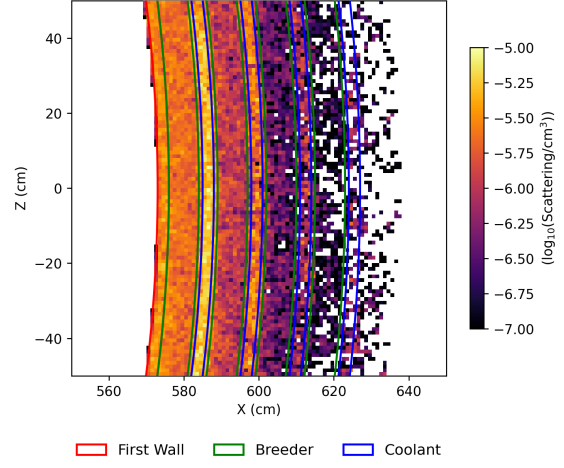
5 Discussion

5.1 Heterogeneity and Blanket Thickness

The data exhibited in figure 3 clearly demonstrates that a heterogeneous blanket results in a significantly reduced TBR compared with a homogeneous blanket with a comparable chemical composition. In a homogeneous breeder the water coolant acts as a moderator, reducing neutron energies to values where the tritium production cross section increases by several orders of magnitude (Brown et al., 2018). Clearly in a heterogeneous configuration this is not the case. Neutrons are already moderated by scattering within the first breeder layer and are much more likely to be absorbed by the seperators or the coolant. The TBR of both homogeneous designs begins to plateau at a total thickness of $\sim 500\text{mm}$, however, they still increase slowly beyond this. Since neutron transport is probabilistic there will still be a very small increase in TBR even at very large thicknesses. Meanwhile as the depth of the heterogeneous blanket increases



(a) Scattering - ζ Graph



(b) Scattering Reaction Density Map

Figure 9: This figure demonstrates the change in scattering with values of ζ on left and the change in scattering density in a 2D slice on the right.

beyond 500mm the TBR completely levels and even begins to decline slightly.

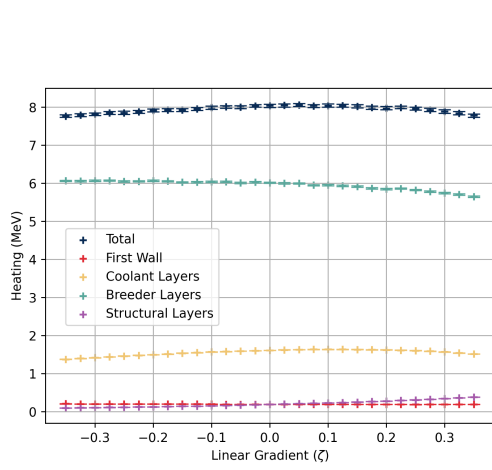
The layered nature of the breeder means that with greater thickness the coolant layer between the first two breeders increases, leading to greater absorption and a reduction in the tritium production of layers 2, 3 and 4. This can be seen clearly in figure 4. With greater thickness the contribution of breeder layer 1 increases linearly. Meanwhile, that of the second layer peaks at ~ 500 mm before declining steadily. Tritium breeding in layers 3 and 4 decline consistently with total thickness. This behaviour is to be expected as thicker front layers reduce neutron transport to the rear of the blanket.

5.2 Neutron Source

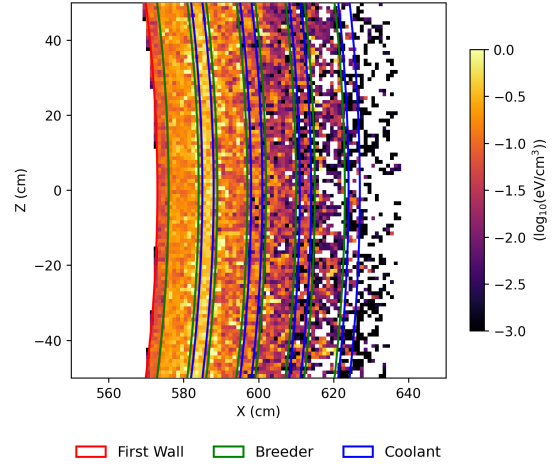
5.3 Linear Gradient

5.3.1 Tritium Breeding Ratio

A linear gradient was imposed on the blanket layers using the factor ζ , allowing for the investigation of neutronic behaviour in fron-loaded and rear-loaded blankets. Tritium breeding plotted in figure 6a remained relatively constant at ~ 1.05 over the range of ζ values. As we would expect, breeder layer 1 contributed the largest TBR in more fronloaded configurations but declined steadily with increasing ζ . TBR in layer 2 remained relatively constant across ζ values while it increased steadily in layers 3 and 4. At identical breeder volumes ($\zeta=0$) the second layer TBR is higher than the first. This we can understood by considering moderation. Neutrons absorbed in the first layer have undergone less scattering than those incident on layer 2 and thus have higher energies. This results in a lower tritium production cross section and a reduced TBR for the same volume. A visualisation of TBR reaction density can be seen in figure 6b for $\zeta=0$. Once again it shows that the TBR of the first two layers is much higher than the rear two. Outside of the breeder layers there is a small but ultimately negligible contribution to TBR due to tritium production reactions involving zirconium in the first wall and deuterium in the coolant. The presence of these reactions were discussed in the first semester report (Peters, 2025) and can be ignored due to their low reaction cross sections (Brown et al., 2018). Overall

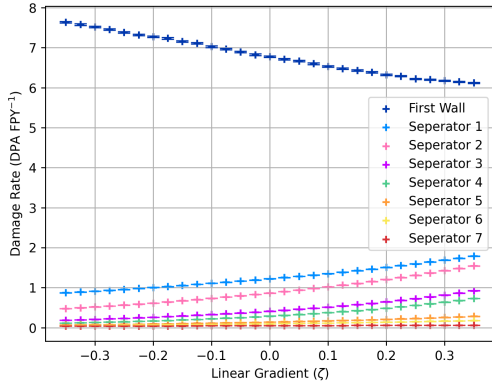


(a) Heating - ζ Graph

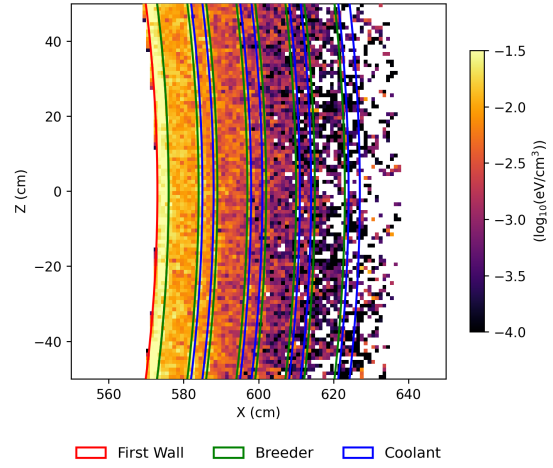


(b) Heating Energy Densit Map

Figure 10: In this figure the graph on the left shows the heating energy in MeV deposited in the breeder structures at values of ζ . The heating energy density in a 2D slice of the breeder is shown on the right.



(a) dpa - ζ Graph



(b) Damage Energy Density Map

Figure 11

the tritium breeding behaviour is in line with our understanding of neutron transport, reflecting the changes in the breeder volume in each layer along with greater neutron penetration into the rear layers at higher values of ζ .

5.3.2 Neutron Multiplication

Neutron multiplication is particularly biased towards the front layers, even more so than TBR. This is clearly visible in figure 7a where the multiplication declines consistently with increasing ζ and in figure 7b where the density of multiplication reactions is almost negligible beyond layer two. Tritium breeding is not only possible but is more likely at thermal neutron energies. The same cannot be said for the neutron multiplication reactions of lead isotopes where there is a ~ 7.5 MeV energy threshold (Brown et al., 2018). Thus, the rate of neutron multiplication declines significantly beyond the second breeder layer as they lose their energy in scattering reactions within the first breeder and coolant layers. A contribution to the total is also recorded by the zirconium in the first wall. This value remains constant across ζ

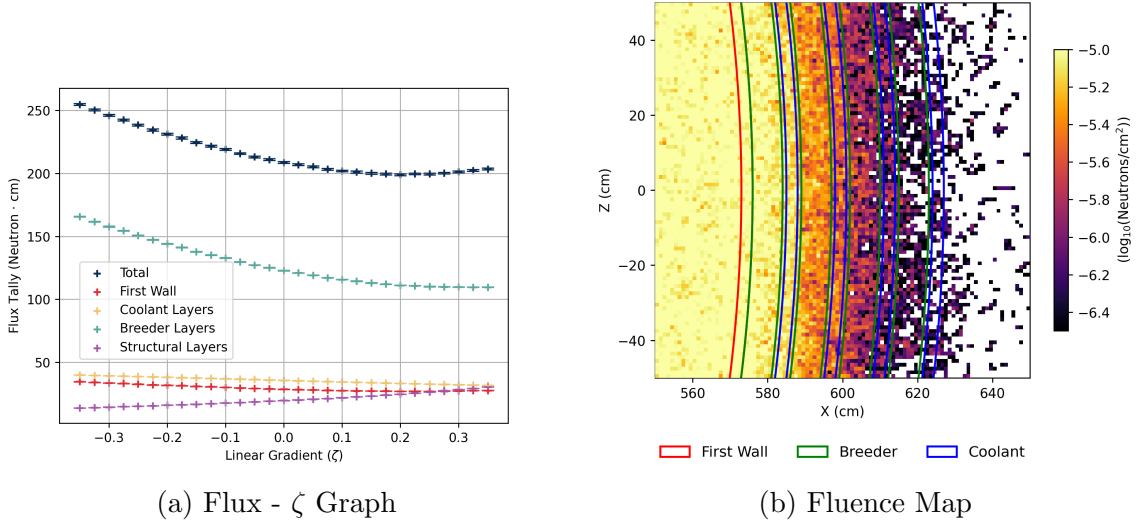


Figure 12

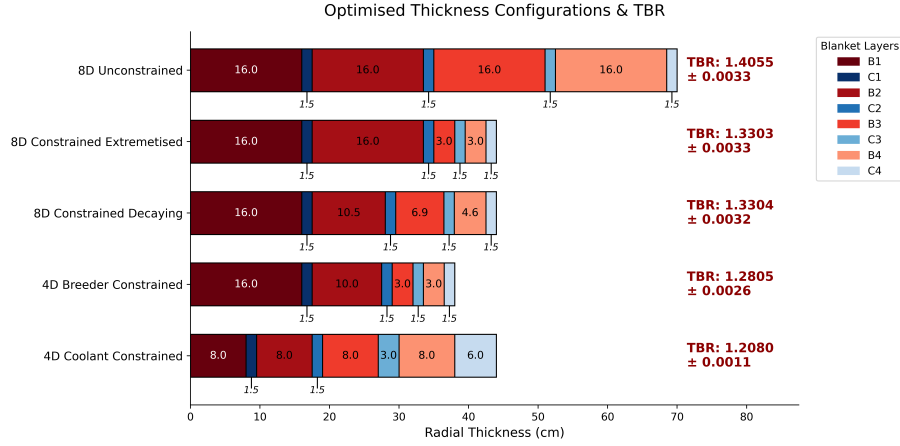


Figure 13: A graph that exhibits the optimum blanket configurations for several 4D and 8D ML simulations and their resulting TBR.

since the reaction has a threshold energy of $\sim 12\text{MeV}$. None of the neutrons produced in the first breeder layer by lead isotope multiplication that are scattered back towards the first wall have sufficient energy to incite multiplication in the zirconium. In short, multiplication in the first wall is solely the product of direct source neutrons, thus the value remains constant as blanket geometry is varied. Meanwhile the $(n,2n)$ reactions that take place in the separator layers by interacting with iron nuclei gradually increase as the blanket becomes more rear-loaded. A thinner first breeder layer results in a greater number of more energetic neutrons interacting with the separators, increasing the multiplication rate.

5.3.3 Neutron Absorption

There are two neutron absorption processes of note that take place in a breeder blanket, the absorption of neutrons that results in tritium production and parasitic absorption. Breeder blanket design is aimed at increasing the former and keeping the latter to a minimum. As depicted in both figure 8a and 8b, the largest contribution to absorption takes place in the breeder layers as neutrons are absorbed to release tritium. Consequently, this value remains relatively constant over ζ , matching the TBR trend demonstrated in figure 6a. The largest

contribution to absorption outside of the breeder material occurs in the Eurofer97 separators. This parasitic absorption decreases with increasing ζ . Thinner first layers result in greater neutrons travelling further and interacting with the separator material at higher energies. Iron isotopes have lower absorption cross sections at higher incident neutron energies (Brown et al., 2018), leading to the decrease in absorption. Very little absorption takes place in the first wall of the breeder despite its proximity to the neutron source. This is due to the relative neutronic transparency of zirconium. Thus, it is no surprise that zirconium is a particular favourite among first wall candidate materials.

5.3.4 Neutron Scattering

Neutrons travelling through the blanket structure can be scattered off nuclei several times before they are absorbed. As ζ increases in figure 9a, the total scattering rate declines before plateauing and gradually increasing at $\zeta > 0.2$. Much like the decline in absorption, this is likely due to the decline in multiplication and the commensurate decrease in neutron population. The contributions to the total from the coolant and breeder layers are relatively similar despite the much greater volume of the breeder. Figure 9b shows the much higher scattering density that takes place in the first coolant layer, highlighting the superior moderation properties of water. The small increase in scattering at high values of ζ is a result of the separator layers. Thinner front layers result in greater penetration and higher energy neutrons interacting with the separator. In contrast to its absorption cross sections, the scattering cross section of iron isotopes is much larger at higher energies. This accounts for the increase in scattering.

5.3.5 Heating Energy

Incident neutrons release energy into a material during inelastic scattering and certain absorption reactions. The total heating displayed in figure 10a remained relatively constant at around 8MeV per source neutron. Most heating occurred in the breeder layers as a combined result of a high scattering rate and the energy released during the exothermic Li_6 tritium breeding reaction (equation 1). The high scattering cross section and low mass of the hydrogen in the water of the coolant layer allows it to extract a significant amount of energy from incident neutrons, resulting in the next highest contribution to the overall heating. In general, it is preferable that these materials, particularly the coolant, be subject to the majority of heat production since they are liquids that can be circulated to distribute heat more effectively.

5.3.6 Damage Energy and Lattice Displacement

In addition to heating the material, energy deposited by neutrons can displace atoms from the crystal lattice. Using equations 3 and 4. The dpa imparted to the first wall and each separator in a full power year is recorded in figure 11a. Since the breeder and coolant are both liquids and not vital to the structural integrity of the blanket they were ignored during this section. As can be seen in figure 11a and 11b the first wall receives the vast majority of damage energy and resulting dpa. This is as we would expect due to its proximity to the neutron source. The dpa fpy^{-1} received by the first wall decreases from 7.52 ± 0.02 to 6.18 ± 0.01 across the range of ζ values. Higher ζ leads to less neutron multiplication in the first layer and less neutrons scattered back into the first wall and less damage. On the other hand, higher ζ allows neutrons to penetrate further into the blanket increasing the damage received by the separator layers.

Material irradiation can result in a wide variety of processes that have complex effects on the properties of reactor components. Helium induced embrittlement resulting from transmutation reactions is the method by which we will characterise material failure. Gilbert et al., 2012 provides DPA thresholds for a variety of materials and grain sizes. For the zirconium in the zircaloy-4 first wall with a grain size of $5\mu\text{m}$ (Woods, 1966) the critical helium embrittlement occurs at a dpa of 61.99. This would suggest a first wall lifetime of 8.23 ± 0.02 to 10.03 ± 0.02 full power years, significantly lower than the 30 years planned for ITER (Aymar, Barabaschi and Shimomura, 2002). While zirconium based materials are neutronicallly superior they are clearly inadequate from a materials stability standpoint. Performing a similar investigation with a Eurofer97 first wall resulted in a $\sim 60\%$ higher absorbed damage energy but a significantly lower dpa fpy^{-1} of between 2.18 ± 0.01 and 2.51 ± 0.01 . Stornelli et al., 2024 describes Eurofer97 with grain size of $0.5\mu\text{m}$. A threshold dpa of 57.47 results in a full-power lifetime of 22.90 ± 0.05 to 26.41 ± 0.06 , much closer to what would be required of a first wall in a functional fusion reactor. Unfortunately, the Eurofer97 first wall reduces the TBR by 9%, below the threshold of 1.05 required for reliable tritium breeding. This underlines a pressing issue in fusion research, where material properties place severe constraints on design and compromises are necessary on neutronic performance may be necessary to guarantee safety.

5.3.7 Neutron Flux

Flux is measured by tracking neutron paths through the breeder structures. Figure 12a shows that the total neutron track length calculated by OpenMC's 'flux' tally declines as ζ increases before reaching a plateau at $\zeta=0.2$. This is likely a result of the reduction in multiplication rate demonstrated in figure 7a. A lower neutron population leads to less neutron travel per source neutron. The breeder layers serve as the main contributor to the total flux as the largest elements by volume and the source of most neutrons born from multiplication reactions. Breeder, coolant and first wall flux all decline steadily across the range of ζ values. However, the neutron track length measured in the separator layers increases linearly as the first breeder layer becomes thinner, allowing greater neutron penetration into the structure. This confirms the scattering and damage energy behaviour discussed previously, where greater neutron transmission depth and incident energy led to increases in these values. The fluence in a 2D segment of the breeder is visualised in figure 12b. As expected the fluence decays exponentially with depth into the blanket, obeying neutron behaviour described by the mean free path.

5.4 Machine Learning and TBR Optimisation

As could be expected the 8-dimensional, unconstrained algorithm resulted in the highest TBR of the trialled ML models with a value of 1.4055 ± 0.0033 . Breeder thicknesses were maximised at 160mm and coolant layers minimised at 15mm. While this undoubtedly results in a higher TBR a blanket of this size would be inefficient and costly to implement. In an attempt to remedy this issue the total blanket thickness (not including separator layers) was limited to 440mm. With this constraint ML model produced a 'decaying' configuration with minimised coolant thicknesses and breeder thicknesses of 160mm, 105mm, 69mm and 46mm. The decreasing breeder thickness led to a TBR of 1.3304 ± 0.0032 . This configuration and the resulting TBR are remarkably similar to the 'binding' blanket design which selects the maximum

value for the first two breeder layers and the minimum value for the rear layers. These results demonstrate clearly that the first two layers are prioritised in terms of tritium breeding, which is in agreement with our results using the linear gradient factor (ζ). Additionally, while the presence of water increased the TBR in a homogeneous breeder, the ML algorithm prioritises the reduction of coolant volume.

When limited to 4-dimensional investigations the peak TBR is appreciably lower. Allowing only either the breeder layers or the coolant layers to vary resulted in tritium breeding ratios of 1.2805 ± 0.0026 and 1.2080 ± 0.0011 respectively. Much like the constrained 8D simulations the 4D breeder model recreated the frontloading phenomena. Conversely, the 4D coolant simulation produced a configuration with coolant layers increasing in thickness further into the blanket. Once again, we see that the tritium breeding effects are concentrated in the first two layers. When limited to a set total depth the model prioritises these layers above the rear two. These insights may be useful when designing a blanket that requires a specific overall coolant flow or has a maximum total depth.

6 Conclusion

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